

HireStream Matching Engine

v2 · STAGING

Logic & Specification — Government of Himachal Pradesh · HPSEDC Overseas Placement Portal

1 Purpose

The HireStream Matching Engine produces a **0-100 fit score** for every candidate ↔ job pair. The score drives the candidate's *Recommended Jobs*, the agent's ranked applicant list, and the match badge on every application. Every score comes with a per-factor breakdown so candidates and agents can see *why* — the algorithm is fully explainable. All weights, missing-criteria policies and the recommendation threshold are admin-tunable at runtime via the Matching Engine Parameters module (§7) — no code redeploy needed.

2 Scoring Framework

Total **100 points** across **7 factors** in three tiers — Hard criteria (60) gate the match, Preferences (25) rank it, Compatibility (15) breaks ties.

TIER	FACTOR	WT	WHAT IT MEASURES
Hard criteria (60 pts)	Skills match	30	Overlap between job's required skills and candidate's listed skills
	Experience	20	Candidate meets job's years-of-experience requirement
	Qualification	10	Candidate's education level meets the job's stated minimum
Preferences (25 pts)	Country	15	Job's destination is in candidate's preferred countries list
	Language proficiency	10	Candidate meets job's language requirement (CEFR or IELTS)
Compatibility (15 pts)	Job category	10	Job's role category is in candidate's preferred categories
	Salary expectation	5	Job salary falls within candidate's expected range
TOTAL		100	Final match score (0-100)

3 Per-Factor Logic

Skills match

30 pts

```
round((matched / required) × 30)
```

Case-insensitive overlap of `job.skills[]` against `candidate.skills[]`. Job with no skills listed → neutral 15.

Experience

20 pts

```
round(min(have/required, 1) × 20)
```

Candidate's years vs job's required. Capped at 100% — extra experience not penalised. `required=0` → full 20.

Qualification

10 pts

```
school < diploma < bachelor < master < doctorate
```

Meets/exceeds: 10. One tier below: 5. Two+ below: 0. No requirement on job: 10.

Country preference

15 pts

```
job.country ∈ candidate.preferred ? 15 : 0
```

Hard match against the candidate's preferred destinations. Empty candidate prefs → neutral 7.

Language proficiency

10 pts

```
prorated | CEFR for non-English | IELTS band for English
```

Each required language scored individually then averaged. **For IELTS countries** (UK / AUS / NZ / CAN / IE), job stores required IELTS overall band (e.g. 6.0); candidate stores their actual band. Meets/exceeds = full marks, one band/level below = half, more below = 0. No requirement → 10.

Job category

10 pts

```
job.category ∈ candidate.preferred ? 10 : 0
```

Controlled vocabulary (Factory Worker, Construction, Driver, Electrician, Plumber, Helper, Hospitality, Caregiver, Healthcare, IT, ...). Empty candidate prefs → neutral 5.

Salary expectation

5 pts

```
within range → 5 pts | ≤10% below expected → 3 pts | outside → 0 pts | either side blank → 5 pts (neutral)
```

Compares `job.salary` against the candidate's expected range (min/max, currency-normalised). Lowest-weighted because salary in overseas placement is often negotiated post-shortlist.

Defaults are baseline only. All seven weights + missing-criteria policies + the recommendation threshold are stored in `system_settings.matching.*` and exposed to admin via the Matching Engine Parameters module (§7). Engine reads the live config on every score request — tuning takes effect immediately.

4 Missing-Criteria Behaviour

What happens when a job doesn't specify a criterion (e.g. no language requirement), or a candidate hasn't filled a field (e.g. no salary expectation)? The engine applies a per-factor "**missing → neutral**" policy. Defaults below; admin can override each cell from \$7.

How to read each cell: policy → points awarded out of that factor's weight (shown in parentheses next to the factor name). E.g. **Half → 15 pts** means the Half policy awards 15 of Skills' 30 points. The arrow is "results in", not multiplication.

FACTOR	JOB-SIDE MISSING	CANDIDATE-SIDE MISSING	BOTH MISSING	RATIONALE
Skills (30)	Half → 15 pts	Zero → 0 pts	Half	Job-blank: generous (employer was sloppy). Candidate-blank: 0 — no skills to match against a job that needs them.
Experience (20)	Full → 20 pts	Zero → 0 pts	Full	Job-blank (0 yrs required): no constraint. Candidate-blank = 0 yrs, scored 0 against any positive requirement.
Qualification (10)	Full → 10 pts	Half → 5 pts	Full	Encourages profile completion without blocking
Country (15)	n/a*	Half → 7 pts	Half	Empty prefs ≠ disinterest — neutral instead of zero
Language (10)	Full → 10 pts	Zero → 0 pts	Full	Gaps must surface — candidate hasn't claimed proficiency
Category (10)	n/a*	Half → 5 pts	Half	Candidate may have wide interests
Salary (5)	Full → 5 pts	Full → 5 pts	Full	Salary often disclosed post-shortlist — don't penalise either side

Both-missing rule: the engine checks the *job side first*. If the job didn't specify a criterion it is treated as "no constraint" and the *Job-side missing* policy applies — so when both sides are blank, the job-side value wins. Each direction is independently tunable per factor from the admin Parameters Module (\$7): *Job didn't specify* and *Candidate didn't provide* are separate Full / Half / Zero knobs.

* **n/a** = that side is a *required field* on the job (Country, Category) so it can never actually be blank — the job-side-missing case simply cannot occur. Every factor still counts toward the fixed 100-point total; missing inputs earn the policy value above (they are *not* dropped from the denominator).

Worked example. Job: Software Engineer in UAE, skills [React, Node], 3y experience — no qualification / languages / salary specified.

Candidate: skills [React, Node, Python], 5y, preferredCountries=[UAE], no qualification / categories / salary set.

Score: Skills 30 + Exp 20 + Qual 10 (job missing) + Country 15 + Lang 10 (job missing) + Cat 5 (cand half) + Salary 5 (both missing) = **95/100**. A partially-filled profile against a partially-filled job still yields a meaningful, defensible score.

5 Threshold, Sort & Explainability

Recommendation cut-off

The candidate's Recommended Jobs view shows only jobs scoring $\geq$ the configured threshold. Default **50**; admin adjusts at runtime via `matching.recommendation_threshold_pct` (\$7).

Sort & tie-break

Default sort on agent's per-job list: match score descending; ties broken by application `appliedAt` (most recent first). Agent can override via dropdown.

Explainability

Every score is returned with a structured per-factor breakdown (score + max + plain-English detail) that the UI renders inline on every match.

Why it matters: HPSEDC is a government scheme — candidates have a right to know why they were or weren't matched. Showing the breakdown avoids grievance escalations and gives agents data to coach candidates ("Add Hindi → you'd jump from 65 to 72").

6 Implementation

Where it lives

- `calculateScoreBreakdown(candidate, job)` in `server/routes/application.routes.ts`
- Cached on `applications.matchScore` at apply-time; recomputed on candidate profile change (toggleable setting).
- Engine version exposed via `/api/v1/matching/version`.

Backward compatibility

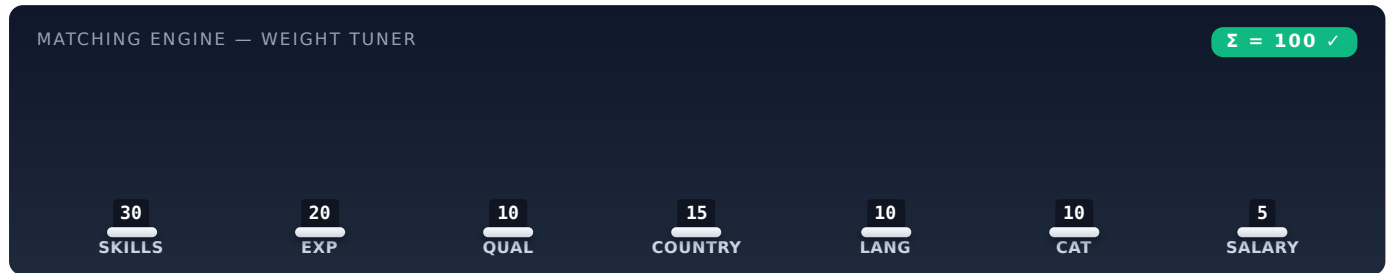
- v2 adds four new factors. Each defaults per \$4 when its input is missing — v1-era profiles don't lose points unfairly.
- Old scores stored in `applications.matchScore` stay valid until next recompute.
- v2 rolls out without a one-time backfill.

7 Matching Engine Parameters Module (Admin)

Every parameter the engine uses is admin-tunable from a dedicated **Matching Engine** tab in the Admin Console. No code change required — HPSEDC ops can fine-tune as the candidate pool, job mix and feedback loop evolve.

7.1 • Weight tuner — "music equalizer" UI

Seven vertical sliders, one per factor. Move a slider to change its weight; the engine reads the live config on every score request. Sum-to-100 enforced on save.



7.2 • Missing-criteria policy

Each cell in the §4 table is an admin-editable dropdown (**Full** / **Half** / **Zero**). Useful for tightening matching as the candidate pool matures — e.g. flip *Candidate-side missing* → *Qualification* from Half to Zero once profile-completion drives are done.

7.3 • Threshold & engine toggle

- **Recommendation threshold:** slider 0–100, default 50
- **Engine version:** v1 / v2 radio (rollback safety)
- **Recompute on profile change:** on/off
- **Show breakdown to candidate:** on/off

7.4 • Live preview

Admin picks a candidate + a job from dropdowns; the breakdown renders live as weights are dragged. Lets HPSEDC verify the impact of a tuning change *before* hitting Save — "If I drop Skills from 30 to 25, does Arjun Sharma still pass the threshold for AI Engineer?"

7.5 • Audit trail

Every weight, policy or threshold change writes to `audit_log` with `actor`, `from`, `to`, `timestamp` and an optional `reason` note. Admin sees the full history in a table within the same module — defensible if a candidate ever disputes their score.

Operational guardrail. Weights must sum to exactly 100. Admin UI rejects any non-summing change with the message *"Weights must sum to exactly 100 (current sum: X)."* A "Reset to defaults" button restores the v2 baseline.

8 Not in v2 (Future Roadmap)

- **Soft skills weighting** (collaboration, leadership) — requires text analysis; deferred until extractor in place.
- **Employer reputation** — placement success rate per employer feeding back into the score; needs ≥ 1000 placements.
- **Freshness decay** — score halves over 30 days; encourages active candidates.
- **ML-driven re-rank** — learn optimal weights from (features → placement_success) once we have enough outcomes data.